Learning to Move Before Learning to Do: Task-Agnostic pretraining for VLAs

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1 Why current VLA training paradigms bottleneck scaling?

- **Scarcity of expert demonstrations:** Models heavily rely on human teleoperation—expensive triplets of observations, language instructions, and actions, making data collection prohibitively expensive.
- **Passive and non-scalable acquisition:** Robots act merely as vessels for human demonstrations without spontaneous exploration, limiting the diversity required for general-purpose manipulation.

Our core idea:

Inspired by infant learning, learning "how to move" can be decoupled from "what to do" using abundant task-agnostic data.

‣ We introduce **TAP (Task-Agnostic Pretraining)**, an innovative two-stage framework that extracts robust physical priors from cheap, unlabeled interaction data before task-specific alignment.

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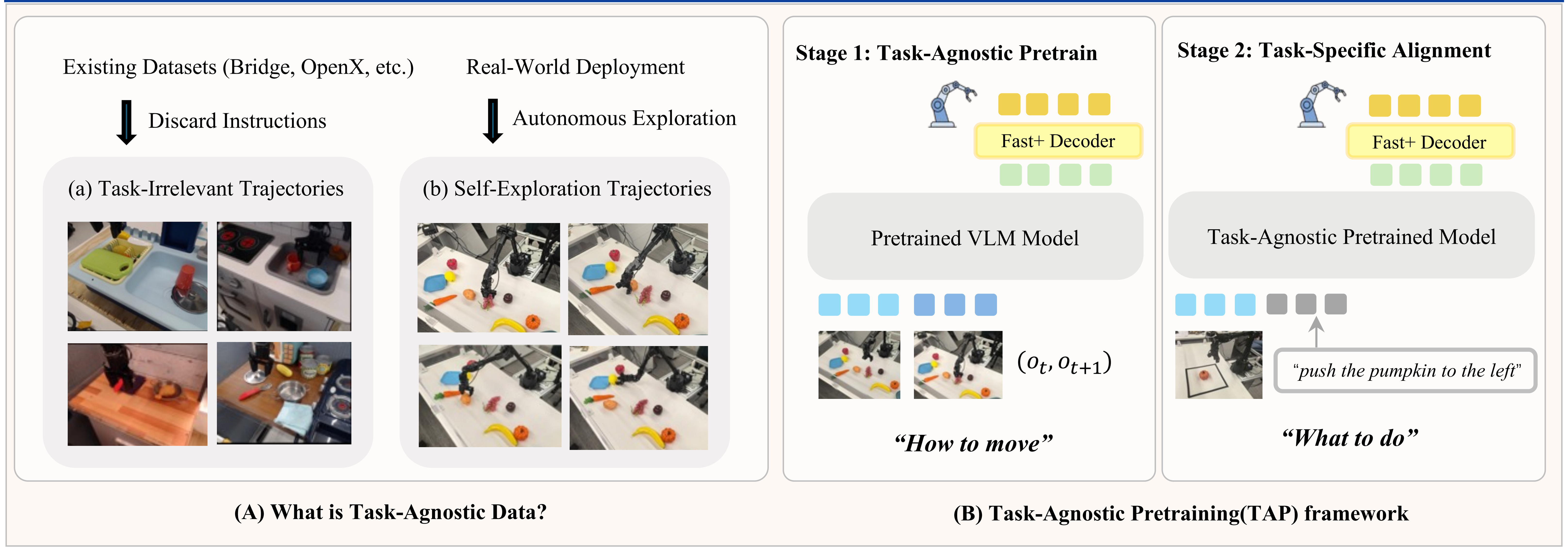


(A) What is Task-Agnostic Data? (B) Task-Agnostic Pretraining(TAP) framework

2 Key Contribution

- **The Decomposition Hypothesis**: We formalize that low-level affordances ("how to move") can be learned without language supervision, reserving expert data for high-level semantics.
- **Task-Agnostic Pretraining (TAP):** A novel two-stage framework leveraging Inverse Dynamics on discarded/play data to instill physical common sense.
- **Exceptional Data Efficiency**: Matches foundational models trained on 1M+ expert trajectories (e.g., OpenVLA, NORA) using orders of magnitude less labeled data (achieving a 10% absolute gain over standard BC).
- **Zero-Shot Robustness**: Pretraining on diverse exploration data yields domain-invariant representations, maintaining strong performance under severe visual shifts where internet-scale baselines catastrophically fail.

4 Experiment Results

Simulation Experiment (Simpler)

Type	Model Name	Spoon on cloth		Carrot on plate		Stack Blocks		Eggplant in Basket		Avg-Partial	Avg-Entire	Avg-All
		Part.	Ent.	Part.	Ent.	Part.	Ent.	Part.	Ent.			
Reference	RT-1-X	4.2%	0.0%	16.7%	0.0%	0.0%	0.0%	3.3%	0.0%	6.05%	0.00%	3.03%
	OpenVLA	4.1%	0.0%	33.0%	0.0%	12.5%	0.0%	8.3%	4.1%	14.48%	1.03%	7.75%
	Nora	37.5%	16.7%	48.0%	0.0%	41.7%	12.5%	4.17%	0.0%	32.84%	7.29%	20.06%
	Octo	**50.0%**	33.0%	**50.0%**	**25.0%**	29.2%	0.0%	**40.0%**	**23.3%**	42.30%	20.33%	31.31%
	π_0	45.8%	29.1%	25.0%	0.0%	50.0%	**16.7%**	**91.6%**	**62.5%**	**53.10%**	**27.05%**	**40.08%**
Baseline	Standard BC	41.7%	33.3%	48.0%	8.0%	37.5%	**16.7%**	0.0%	0.0%	31.79%	14.50%	23.15%
Ours	**-8k episodes**	**50.0%**	**37.5%**	37.5%	8.3%	**58.3%**	4.2%	0.0%	0.0%	36.45%	12.50%	24.47%
	-14k episodes	41.7%	33.3%	**50.0%**	**16.7%**	**83.3%**	12.5%	4.2%	0.0%	44.80%	15.62%	30.21%
	-20k episodes	**66.7%**	**58.3%**	**50.0%**	0.0%	**58.3%**	**16.7%**	8.3%	8.3%	**45.82%**	**20.82%**	**33.32%**

Real-World Experiment (WidowX)

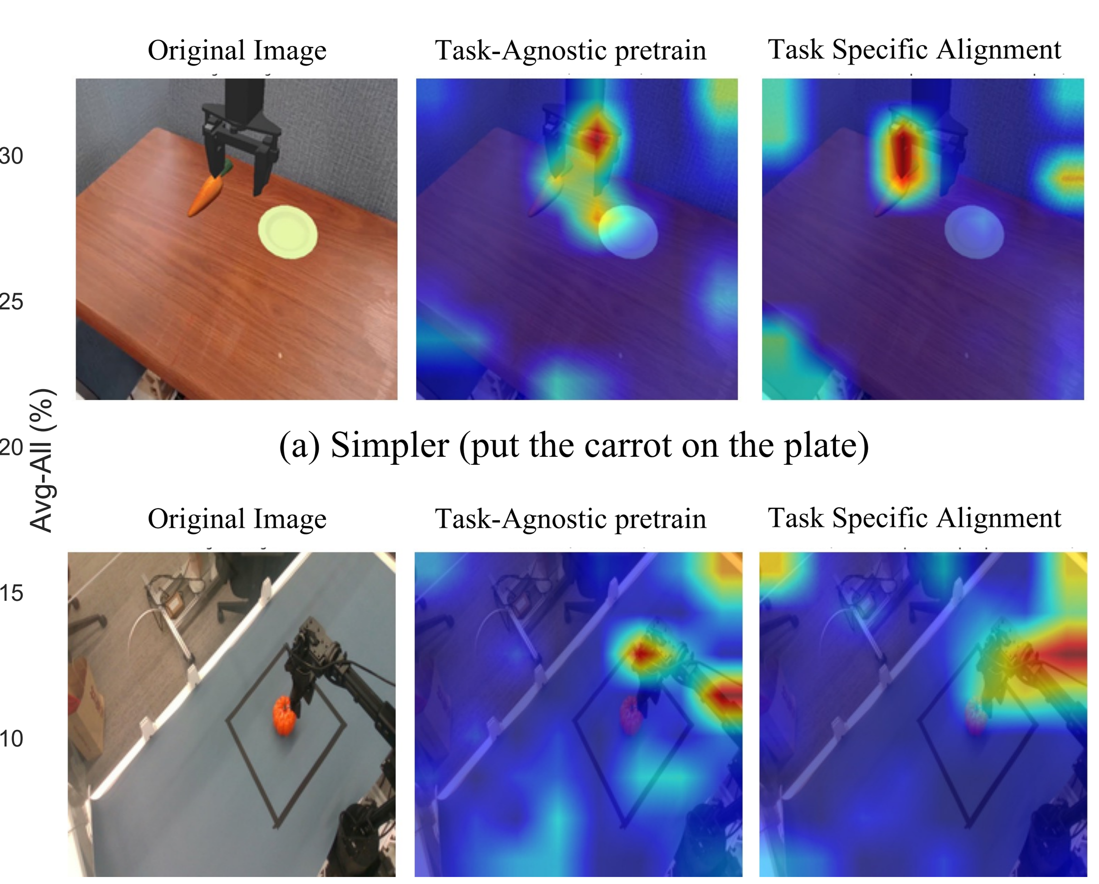
Evaluation Scenario	Task: Put the carrot on the plate			Task: Push the pumpkin to the left		
	From scratch	**(Ours)**	NORA (SOTA)	From scratch	**(Ours)**	NORA (SOTA)
Standard Setup	20%	40%	**65%**	55%	75%	**85%**
Initial State Perturbation	20%	30%	**65%**	45%	75%	**80%**
Visual Distractors	5%	30%	**40%**	5%	**65%**	60%
Background Texture Shift	0%	**25%**	10%	0%	**65%**	55%
Viewpoint Variation	0%	**15%**	0%	0%	**25%**	0%
Average	9%	28%	**36%**	21%	**61%**	56%

- TAP achieves exceptional data efficiency, outperforming internet-scale VLA models in simulation while using orders of magnitude less labeled data.
- Furthermore, real-world deployments demonstrate profound zero-shot robustness, maintaining stable manipulation under severe visual and geometric shifts where traditional baselines catastrophically fail.

5 How Task-Agnostic Data Shapes Learning

‣ **Data Scaling Dependency**: The heatmap shows that downstream performance is strictly bounded by the scale of Stage 1 pretraining. Abundant task-agnostic data acts as a crucial foundation to prevent overfitting during finetuning.

‣ **Emergent Physical Saliency**: Attention maps show Stage 1 implicitly learns physical affordances (focusing on manipulable objects/grippers without text). In Stage 2, language instructions strictly constrain attention for precise motor execution.



(a) Simpler (put the carrot on the plate)

(b) Real World (push the pumpkin to the left, Unseen background)